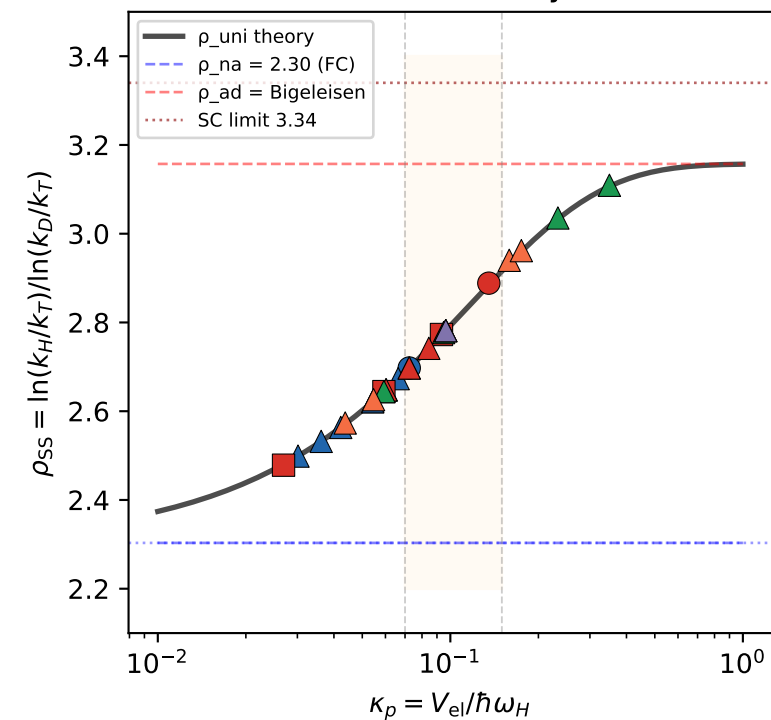
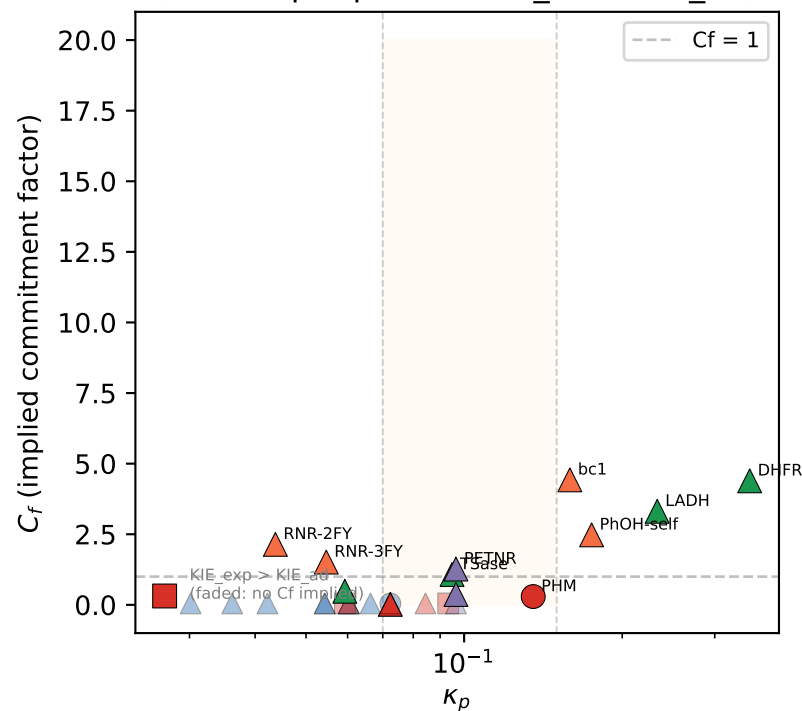


Mechanistic Map: Adiabaticity, Swain-Schaad Exponent, and Commitment Factors across 28 Enzyme PCET Systems

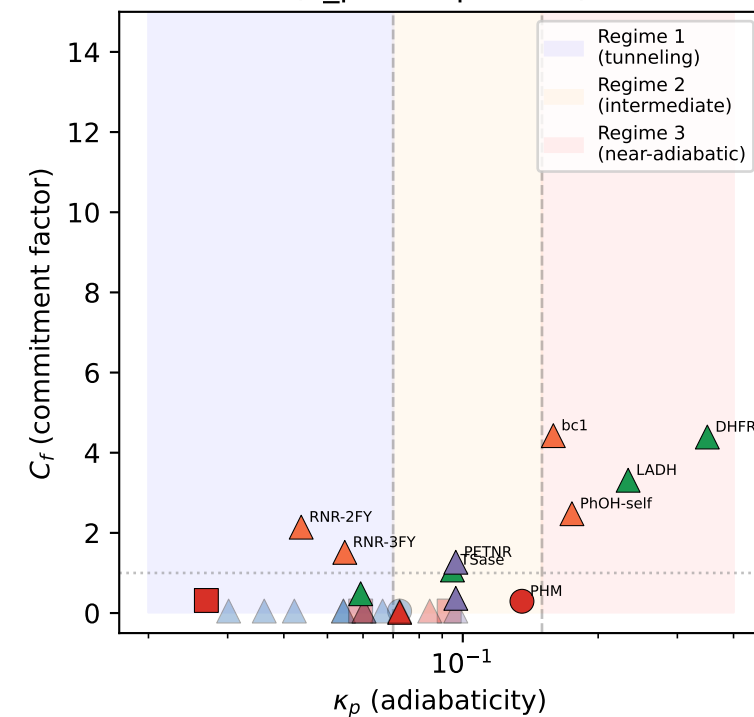
Swain-Schaad exponent vs adiabaticity



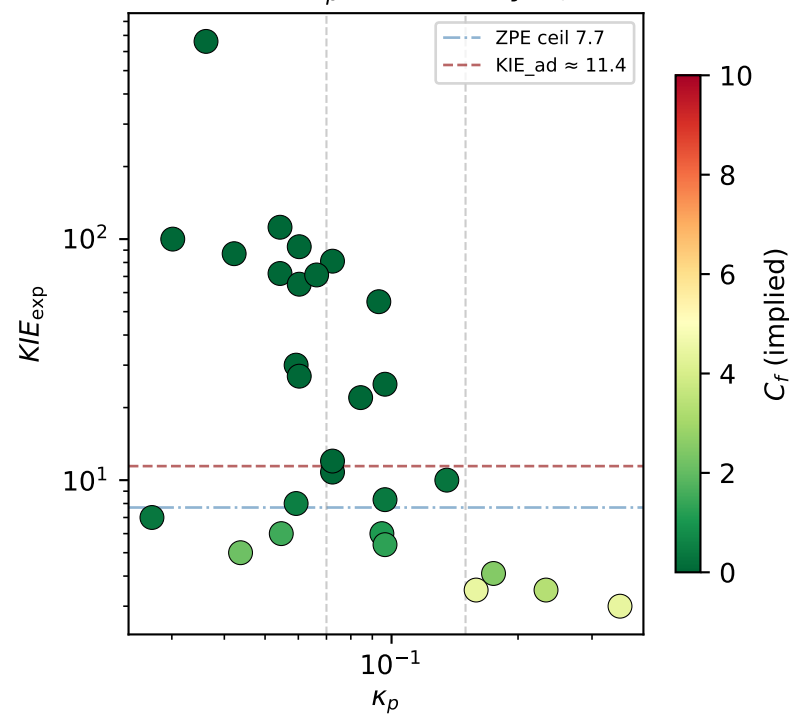
Commitment factor (Northrop equation, $KIE_{int} = KIE_{ad}$)



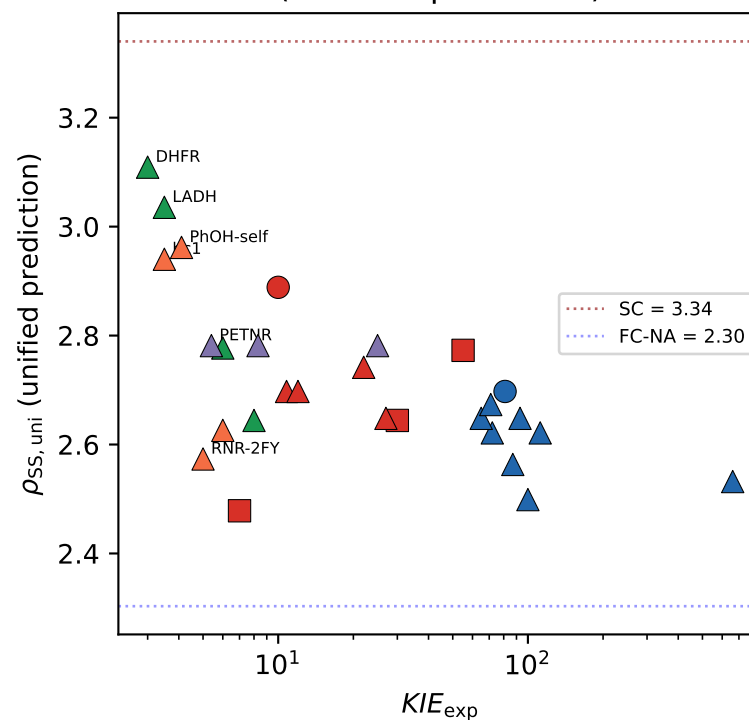
2D mechanistic map (κ_p vs implied C_f)



KIE vs κ_p , shaded by C_f



SS exponent vs experimental KIE (testable prediction)



H, D, T Landau-Zener weights (mass asymmetry drives SS correction)

